

CAPER-AMC: Physics-Enhanced Dynamic Attention Routing for Automatic Modulation Classification

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Abstract—Automatic Modulation Classification (AMC) is important for cognitive radio, spectrum monitoring, and intelligent wireless communication, but reliable classification remains difficult under different signal and SNR conditions. Existing CNN-Transformer and attention-based AMC methods usually apply a fixed attention mechanism or processing path to all received signals. To address this limitation, this paper proposes CAPER-AMC, a physics-enhanced dynamic attention routing framework that combines RF-aware features, multi-scale CNN representations, spectral and analog information, SNR and uncertainty estimation, and adaptive selection among Full, Causal, and Local attention routes. Conditional Top-1 routing executes only the selected route, while specialized experts improve difficult QAM and analog modulation pairs. Experiments on RML2016.10a from -6 to 18 dB achieve 88.56% test accuracy and 0.8801 Macro-F1, improving the 85.05% accuracy of the reference fixed-attention model and demonstrating a better accuracy–efficiency trade-off.

Index Terms—Automatic Modulation Classification, CNN, Transformer, Dynamic Routing, Attention, Radio Frequency, Physics-Aware Learning

I. INTRODUCTION

Automatic Modulation Classification (AMC) is used to automatically identify the modulation type of a received wireless signal. It plays an important role in cognitive radio, spectrum monitoring, signal intelligence, and intelligent wireless communication systems. However, wireless signals are often affected by noise and changing channel conditions, which makes reliable modulation recognition difficult. Therefore, an effective AMC system should achieve high classification accuracy while maintaining low computational cost. Deep-learning-based AMC methods have improved modulation recognition by learning useful features directly from I/Q signals. CNNs are effective for extracting local signal patterns, while Transformer and attention-based models can capture longer temporal relationships. Recent CNN-Transformer and attention-based methods have shown strong performance [1], [4], [11]. However, most existing approaches use a fixed attention mechanism or the same processing path for every input. This may be inefficient because different modulation classes and SNR conditions contain different temporal, spectral, and physical characteristics. Catak et al. [1] also showed that

Full, Causal, and Sparse attention provide different accuracy–efficiency trade-offs. Thus, dynamically selecting an appropriate attention mechanism for each signal remains an important research problem. To address this gap, this paper proposes CAPER-AMC, a physics-enhanced dynamic attention routing framework. The model combines physics-aware I/Q features, multi-scale CNN representations, spectral and analog information, SNR estimation, prediction uncertainty, and adaptive routing among Full, Causal, and Local attention mechanisms. Conditional Top-1 routing executes only the selected attention route for each signal, while specialized experts assist with difficult QAM and analog modulation pairs. Experiments are performed on the RML2016.10a dataset with 11 modulation classes over SNR values from -6 to 18 dB. The proposed method achieves 88.56% test accuracy and 0.8801 Macro-F1, compared with 85.05% accuracy of the reference fixed-attention model.

The main contributions of this work are:

- A physics-enhanced representation using temporal, amplitude, phase, frequency, and spectral signal information.
- A dynamic attention router guided by estimated SNR and prediction uncertainty.
- Conditional Top-1 routing among Full, Causal, and Local attention to improve the accuracy–efficiency trade-off.
- Specialized processing for difficult QAM and analog classes, achieving 88.56% test accuracy on RML2016.10a.

The CAPER-AMC implementation is publicly available at: <https://github.com/Takbir-Al-Tanim/CAPER-AMC>.

II. RELATED WORK

Catak et al. [1] compared baseline, causal, and sparse attention in a CNN-Transformer AMC model, showing an accuracy–efficiency trade-off. Ma et al. [2] proposed DWMTN using a Mamba-Transformer architecture with adaptive feature fusion. Cai et al. [3] introduced efficient convolution and graph sparse attention for lightweight AMC. Ma et al. [4] combined CNN and Transformer modules to capture local and long-range modulation features. Zhang et al. [5] used

recurrent self-attention to extract spatial, frequency, and temporal information. Wu et al. [6] combined complex-valued convolution, Bi-LSTM, and Transformer modules for local and global RF feature learning. Huo et al. [7] proposed SMTrans, a scale-aware Transformer designed for efficient modulation recognition. Zeng et al. [8] developed an SNN-based Transformer to reduce computational requirements for AMC. Wang et al. [9] exploited heterogeneous time, frequency, image, and graph representations for modulation recognition. Ruikar et al. [10] proposed HCTC by combining convolutional feature extraction with Transformer-based temporal modeling. Ansari et al. [11] introduced an attention-enhanced hybrid deep learning–Transformer framework for robust AMC. Xin et al. [12] proposed CPPCNet for high-performance and low-complexity AMC in resource-constrained IoT systems. Wang et al. [13] developed a lightweight temporal hybrid network for robust modulation classification. Zhang et al. [14] investigated multimodal information processing to obtain richer modulation features. Han et al. [15] proposed an RF-retentive network for efficient long-range signal modeling. Finally, Wang et al. [16] combined complex-valued convolution with multi-stream spatiotemporal feature fusion for robust AMC.

A. Research Gap

Table I summarizes the main limitations of existing AMC methods and the corresponding research gaps.

TABLE I
RESEARCH GAP ANALYSIS

Research Field / Related Work	Limitation and Research Gap
Attention & Transformer AMC [1], [2], [4], [5], [11]	Fixed attention is commonly used; signal-dependent attention routing remains limited.
Efficient & Lightweight AMC [3], [7], [8], [12], [13], [15]	Efficiency mainly relies on fixed lightweight models; adaptive accuracy–cost control is needed.
RF-Aware & Multimodal [6], [9], [14], [16]	RF features support classification, but their use for guiding attention routing remains limited.
Hybrid CNN–Transformer [1], [4], [6], [10]	Most models follow a fixed computation path; dynamic local/global routing is insufficiently explored.
Accuracy–Efficiency [1], [3], [7], [8], [12]	Accuracy and computational cost are rarely considered jointly during input-dependent routing.

CAPER-AMC addresses these gaps through RF-guided, cost-aware dynamic attention routing to balance classification accuracy and computational efficiency.

III. METHODOLOGY

A. Dataset

The experiments use the RML2016.10a dataset, a widely used benchmark for Automatic Modulation Classification (AMC). The original dataset contains 220,000 synthetically generated I/Q signal examples. Each sample contains 128

complex-valued signal points represented by in-phase (I) and quadrature (Q) components. The dataset contains 11 modulation classes: 8PSK, AM-DSB, AM-SSB, BPSK, CPFSK, GFSK, PAM4, QAM16, QAM64, QPSK, and WBFM. In this experiment, signals with SNR values from -6 dB to 18 dB are selected, resulting in 143,000 samples across 13 SNR levels. The selected data are divided using a stratified 60%/20%/20% training, validation, and testing split within every modulation-SNR group. A short exploratory data analysis (EDA) is performed before training. The I/Q waveform, instantaneous amplitude, phase variation, frequency change, and magnitude spectrum are examined at different SNR levels. These observations show that amplitude, phase, and spectral information provide complementary information for identifying different modulation types.

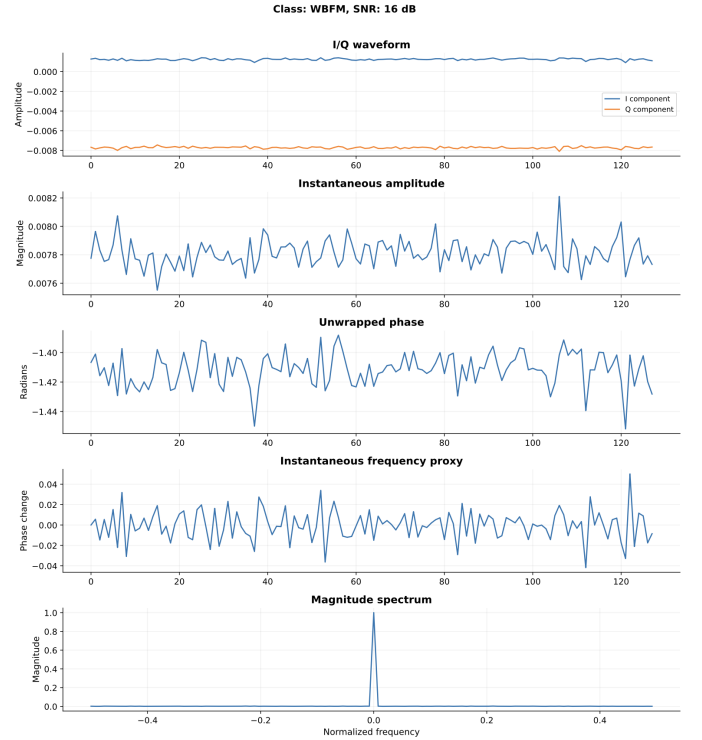


Fig. 1. Example I/Q waveform and physics-derived features for a WBFM signal at 16 dB SNR.

B. Proposed CAPER-AMC Framework

The proposed CAPER-AMC framework dynamically selects an appropriate attention mechanism for each input signal instead of applying one fixed attention pattern to all samples. As illustrated in Fig. 2, the framework combines physics-aware features, spectral and analog context, a multi-scale CNN encoder, SNR- and uncertainty-based routing, and Full, Causal, and Local attention routes. The selected representation is then fused with context information and processed by the main classifier and pair experts for final modulation prediction.

1) *Physics-Aware Input*: The input signal is represented as

$$X \in \mathbb{R}^{2 \times 128}, \quad (1)$$

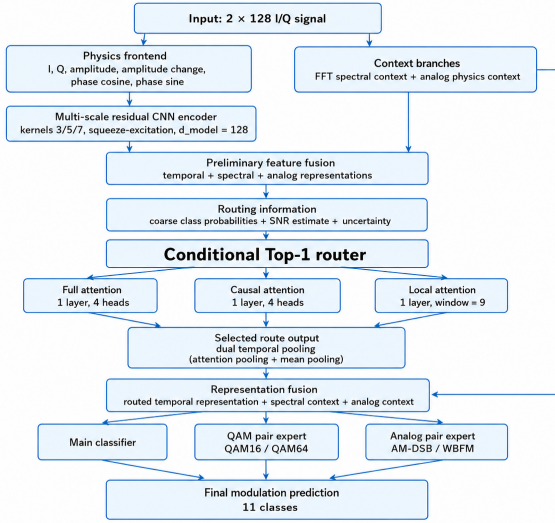


Fig. 2. Overall architecture of the proposed CAPER-AMC framework.

where the two channels represent the I and Q components. Instead of using only raw I/Q samples, the proposed model extracts additional physics-based information. The instantaneous amplitude is calculated as

$$A_t = \sqrt{I_t^2 + Q_t^2}, \quad (2)$$

and the amplitude variation is calculated as

$$\Delta A_t = A_t - A_{t-1}. \quad (3)$$

Amplitude, amplitude change, and phase-motion information are combined with the original I/Q channels to form a six-channel physics-aware input. This representation allows the model to directly observe useful RF signal characteristics.

2) *Multi-Scale CNN and Spectral/Analog Context*: A multi-scale CNN encoder extracts temporal features using convolution branches with kernel sizes 3, 5, and 7. Residual connections and channel attention allow the network to capture short- and medium-range signal patterns. In parallel, frequency-domain information is extracted using the Fast Fourier Transform (FFT):

$$P(f) = \log(1 + |\mathcal{F}(I)|^2 + |\mathcal{F}(Q)|^2). \quad (4)$$

The spectral representation provides global frequency information, while an additional analog context module extracts amplitude, phase, frequency, coherence, and band-energy statistics. These features are particularly useful for distinguishing difficult analog classes such as AM-DSB and WBFM.

3) *SNR and Uncertainty Router*: Temporal, spectral, and analog features are first combined into a preliminary representation. A coarse classifier produces initial class probabilities, while another branch estimates the input SNR. Prediction uncertainty is measured using normalized entropy:

$$U = -\frac{1}{\log K} \sum_{k=1}^K p_k \log p_k, \quad (5)$$

where $K = 11$ is the number of modulation classes. The router receives the preliminary representation, estimated SNR, and uncertainty:

$$z_r = [z_{\text{pre}}, \widehat{\text{SNR}}, U]. \quad (6)$$

The router then produces probabilities for the Full, Causal, and Local attention routes.

4) *Dynamic Full/Causal/Local Attention*: Three attention routes are available in CAPER-AMC. Full attention captures global dependencies between all signal positions. Causal attention restricts each position to current and previous positions, while Local attention focuses only on a limited temporal window of nine samples. Full attention is calculated as

$$\text{Att}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (7)$$

During training, straight-through Gumbel-Softmax is used to learn hard routing decisions. During inference, Conditional Top-1 routing selects only the route with the highest probability:

$$r^* = \arg \max_r p_r. \quad (8)$$

Therefore, only one attention route is executed for each input sample.

5) *Pair Experts and Classification*: After routed attention, the temporal representation is combined with spectral and analog context features. Two specialized pair experts are used for the most difficult modulation pairs. The QAM expert focuses on QAM16 and QAM64, while the analog expert focuses on AM-DSB and WBFM. These experts refine the prediction when the sample is likely to belong to one of these difficult pairs. Finally, the fused representation is passed to the classifier to predict one of the 11 modulation classes.

C. Implementation

The proposed CAPER-AMC model is implemented using PyTorch. The main training and architectural hyperparameters are summarized in Table II. AdamW is used for optimization

TABLE II
CAPER-AMC CONFIGURATION

Parameter	Value
Model dimension	128
Attention heads	4
FFN dimension	384
Attention layers per route	1
Local window	9
Dropout	0.08
Batch size	384
Maximum epochs	65
Learning rate	2.5×10^{-4}
Weight decay	5×10^{-5}
Early-stop patience	10

together with cosine learning-rate scheduling, gradient clipping, and exponential moving average checkpointing. Joint I/Q RMS normalization is applied without subtracting the complex mean. Mild RF-specific augmentation includes carrier phase rotation, small frequency offset, I/Q gain variation, and circular temporal shifting.

a) *Step 1: Training:* During training, each mini-batch is normalized and augmented before physics, CNN, spectral, and analog features are extracted. The model estimates the SNR and prediction uncertainty, and the router selects one of the three attention routes using Top-1 Gumbel routing. The selected route processes the signal, the pair experts refine difficult class predictions, and the model parameters are updated using the combined classification and auxiliary losses.

b) *Step 2: Testing:* During testing, the trained model processes each input using deterministic Conditional Top-1 routing. The final predictions are evaluated using classification accuracy, precision, recall, Macro F1-score, confusion matrix, SNR-level accuracy, inference time, and Expected Calibration Error (ECE).

Algorithm 1 CAPER-AMC Training and Inference

Require: I/Q signal X

- 1: Normalize and augment X
 - 2: Extract physics and CNN features
 - 3: Compute spectral and analog context
 - 4: Estimate SNR and prediction uncertainty
 - 5: Compute Full, Causal, and Local route probabilities
 - 6: Select one route using Conditional Top-1 routing
 - 7: Process signal using the selected attention route
 - 8: Apply QAM or analog pair expert when required
 - 9: Generate the final modulation prediction
 - 10: Update model parameters during training
 - 11: Evaluate prediction during testing
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IV. EXPERIMENTAL SETUP

A. Experimental Design

All experiments are implemented in PyTorch and executed on an NVIDIA Tesla T4 GPU. A fixed random seed of 42 is used for reproducibility. The dataset is divided into training, validation, and test sets using the 60%/20%/20% split described in Section III. Model parameters are learned using only the training set, while the validation set is used for model selection, early stopping, and final calibration. The test set is used only for final performance evaluation. The best exponential-moving-average checkpoint is selected according to validation performance. Two test settings are reported. First, single-view inference is used to measure the normal classification performance and inference efficiency of Conditional Top-1 routing. Second, four-view test-time augmentation (TTA) followed by validation-fitted calibration is used to obtain the final reported classification result.

B. Evaluation Metrics

The classification model is evaluated using accuracy, precision, recall, Macro-F1, confusion matrix, inference time, parameter count, and Expected Calibration Error (ECE). Since CAPER-AMC also estimates the SNR as an auxiliary regression task, Mean Absolute Error (MAE) is used to measure SNR estimation performance. For N test samples, classification accuracy is defined as

$$Accuracy = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i = \hat{y}_i), \quad (9)$$

where y_i and $\hat{y}_i$ denote the true and predicted classes, respectively. For modulation class k , precision and recall are calculated as

$$Precision_k = \frac{TP_k}{TP_k + FP_k}, \quad (10)$$

$$Recall_k = \frac{TP_k}{TP_k + FN_k}, \quad (11)$$

where TP_k , FP_k , and FN_k represent the true-positive, false-positive, and false-negative samples of class k . The class-level F1-score is

$$F1_k = 2 \frac{Precision_k Recall_k}{Precision_k + Recall_k}. \quad (12)$$

Because the dataset contains multiple modulation classes, Macro-F1 is used to give equal importance to every class:

$$F1_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K F1_k, \quad (13)$$

where $K = 11$. For the auxiliary SNR estimation task, MAE is calculated as

$$MAE_{\text{SNR}} = \frac{1}{N} \sum_{i=1}^N |s_i - \hat{s}_i|, \quad (14)$$

where s_i and $\hat{s}_i$ are the true and estimated SNR values. Model confidence is evaluated using ECE:

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |Acc(B_m) - Conf(B_m)|, \quad (15)$$

where B_m represents a confidence bin. Lower ECE indicates that the predicted confidence better matches the actual classification accuracy.

V. RESULTS AND DISCUSSION

A. Overall Performance

The overall CAPER-AMC results are summarized in Table III. The selected model achieves a validation accuracy of 88.37%. Under normal single-view inference, the test accuracy is 88.48%. Four-view TTA increases the accuracy to 88.54%, while calibration produces the final test accuracy of 88.56%. The final Macro-F1 is 0.8801, indicating balanced performance across the modulation classes. *Inference time is measured using single-view Conditional Top-1 inference, whereas the final 88.56% result uses four-view TTA and calibration.

B. Performance Across SNR

Figure 3 presents classification accuracy and Macro-F1 across the evaluated SNR range. Both metrics improve rapidly as signal quality increases. At -6 dB, the model achieves approximately 60.8% accuracy, which increases to 76.9% at -4 dB and 83.4% at -2 dB. At 0 dB, the accuracy reaches approximately 89.6%. From 2 dB onward, the model remains generally between 92% and 94% accuracy. This indicates that most classification errors occur under low-SNR conditions, where noise strongly distorts the modulation characteristics.

TABLE III
OVERALL CAPER-AMC RESULTS

Metric	Result
Validation Accuracy	88.37%
Single-View Test Accuracy	88.48%
Raw 4-View TTA Accuracy	88.54%
Final Test Accuracy	88.56%
Macro Precision	0.8980
Macro Recall	0.8856
Macro-F1	0.8801
ECE	0.0112
SNR MAE	1.905 dB
Trainable Parameters	1.06M
Top-1 Inference Time*	0.0359 ms/sample

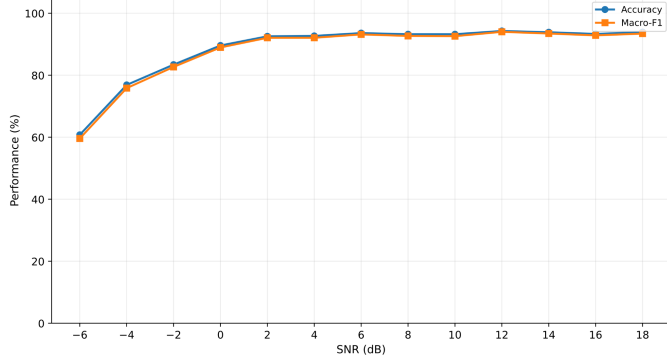


Fig. 3. Accuracy and Macro-F1 versus SNR.

C. Per-Class Performance

Fig 4 shows precision, recall, and F1-score for each modulation class. The model achieves strong performance for most

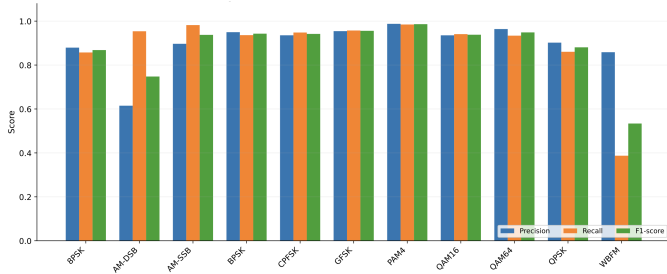


Fig. 4. Precision, recall, and F1-score for each modulation class.

modulation classes. In particular, PAM4 obtains an F1-score of 0.986, GFSK achieves 0.956, QAM16 achieves 0.938, and QAM64 achieves 0.949. These results show strong discrimination of the two QAM classes. In contrast, WBFM remains the most difficult class, with an F1-score of approximately 0.534. This weakness is mainly caused by confusion between WBFM and AM-DSB.

D. Confusion Matrix Analysis

The normalized confusion matrix is shown in Fig. 5. Most classes show a strong diagonal distribution, indicating a high percentage of correct predictions. QAM16 and QAM64

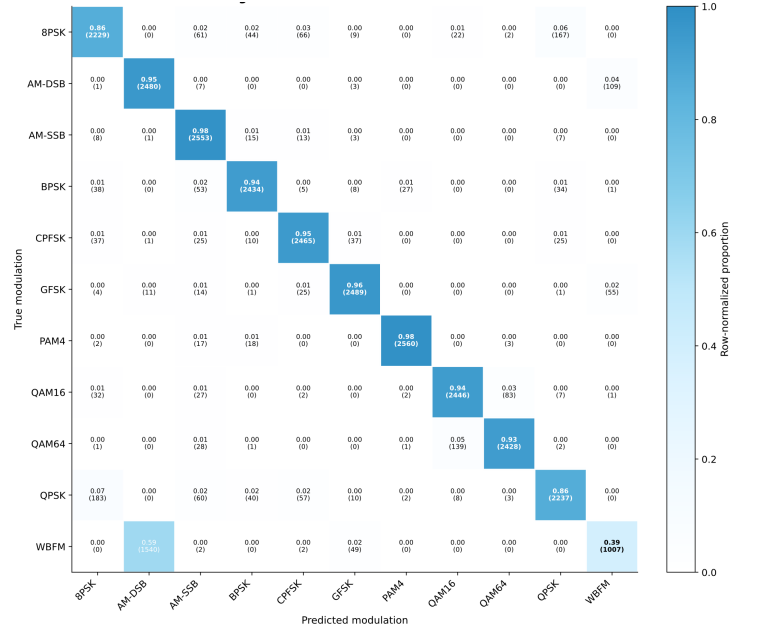


Fig. 5. Normalized confusion matrix of CAPER-AMC.

achieve recalls of approximately 94.1% and 93.4%, respectively. The largest remaining error occurs for WBFM, where many samples are predicted as AM-DSB. This suggests that separating these two analog modulation types remains the main limitation of the current model.

E. Dynamic Routing and Computational Efficiency

The learned routing behavior provides additional insight into CAPER-AMC. Figure 6 shows how route selection changes with SNR. At low SNR, the model selects Local attention

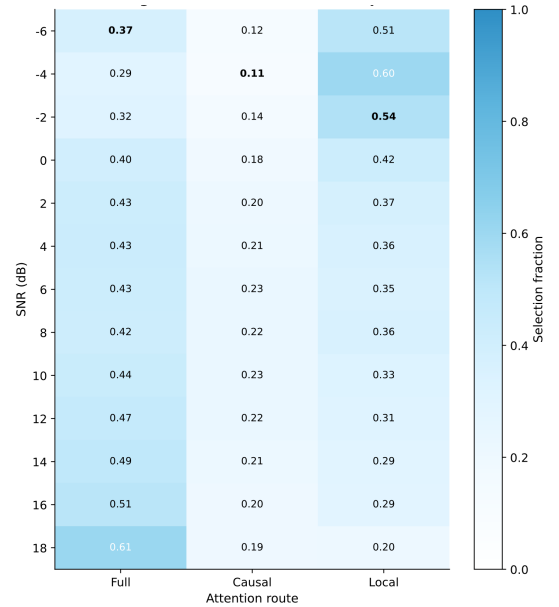


Fig. 6. Selection of Full, Causal, and Local attention routes across SNR levels.

more frequently. For example, at -4 dB approximately 60% of samples use the Local route. As the SNR increases, Full attention becomes more common and reaches approximately 61% at 18 dB. This behavior indicates that CAPER-AMC does not depend on one attention mechanism for every signal condition; instead, it adapts its routing according to the characteristics of the input. Conditional Top-1 routing achieves 88.48% single-view accuracy with an inference time of approximately 0.0359 ms per sample. Processing all attention routes simultaneously produces only a small accuracy increase but requires substantially more inference time. Therefore, Conditional Top-1 provides a useful balance between classification performance and computational cost.

VI. CONCLUSION

This paper presented CAPER-AMC, a physics-enhanced dynamic attention routing framework for automatic modulation classification. The proposed model combines physics-aware I/Q features, multi-scale CNN representations, spectral and analog information, SNR and uncertainty estimation, and adaptive routing among Full, Causal, and Local attention mechanisms. Conditional Top-1 routing allows each signal to use only one selected attention path, while specialized experts improve the classification of difficult QAM and analog modulation pairs. Experiments on RML2016.10a with 11 modulation classes over SNR values from -6 to 18 dB achieved 88.56% final test accuracy and 0.8801 Macro-F1, compared with 85.05% accuracy of the reference fixed-attention model. Single-view Conditional Top-1 inference achieved 88.48% accuracy with an inference time of approximately 0.0359 ms/sample. The learned routing behavior also showed that CAPER-AMC changes its attention selection according to signal conditions, while strong improvements were observed for QAM16 and QAM64. However, performance remains challenging at low SNR and for WBFM, particularly due to confusion with AM-DSB. Future work will focus on repeated-seed validation, improved analog modulation separation, lighter model design, ablation studies, and evaluation under realistic wireless channel conditions.

VII. LIMITATIONS AND FUTURE WORK

Although CAPER-AMC achieves strong overall performance, several limitations remain. Classification performance decreases at low SNR because noise distorts important signal characteristics. WBFM is also frequently confused with AM-DSB and remains the most challenging modulation class. Moreover, the current evaluation is based on a single primary training run, which limits the analysis of model stability. Future work will include repeated experiments with multiple random seeds and independent held-out testing to evaluate robustness and generalization. Further investigation will focus on improving WBFM/AM-DSB separation, reducing model complexity, performing detailed ablation studies, and evaluating CAPER-AMC under realistic fading, multipath, and other wireless channel conditions.

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